

# Mahima Mahabaleshwar Siddheshwar

Bioinformatics Scientist | Genomics & Data Science | Computational Biologist

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## PROFESSIONAL SUMMARY

Bioinformatics Scientist with 3+ years of experience spearheading Next-Generation Sequencing (NGS) analysis, single-cell (scRNA-seq) and bulk RNA-seq to drive biomarker discovery and target identification. Skilled in building reproducible, scalable omics workflows in Python, R, and Bash, with a conceptual understanding of Docker and Nextflow/Snakemake for pipeline containerization. Experienced in leveraging AI/ML agents and statistical modeling to optimize biological reasoning across HPC and cloud environments. Includes hands-on wet-lab exposure to RNA extraction and NGS library prep to support R&D.

## TECHNICAL SKILLS

**Genomic Analysis & Pipelines:** scRNA-seq (Seurat, Scanpy), Bulk RNA-Seq (DESeq2, edgeR, limma), Multi-omics Integration, WGCNA, Variant Calling (GATK), Pathway Enrichment (GSEA, KEGG, Reactome), Spatial Transcriptomics and Omics Data.  
**Bioinformatics Tools & Resources:** STAR, BWA, SAMtools, BEDTools, ANNOVAR, Ensembl, GEO, TCGA, UCSC Genome Browser, Cytoscape, IGV  
**Programming & Data Science:** Python, R, Bash/Shell Scripting, SQL, Git Version Control  
**Machine Learning & Statistics:** PCA, t-SNE, UMAP, K-means, Hierarchical Clustering, LASSO Regression, SVM, Model Evaluation (ROC/AUC)  
**Cloud, HPC & Infrastructure:** HPC Cluster Computing (SLURM), AWS, Google Cloud Platform (GCP), Unix/Linux Workflow Automation  
**Data Visualization & Reporting:** Volcano Plots, Heatmaps, Interactive Dashboards (Plotly, Power BI, Tableau)  
**Wet Lab & Quality Control:** Cell Culture, DNA/RNA extraction, PCR, SDS-PAGE, FACS-based workflows, and NGS library preparation  
**Clinical, Quality & Regulatory:** HIPAA, GxP/GLP, ALCOA+, FAIR Principles, LIMS, FDA Documentation & Audit Readiness

## PROFESSIONAL EXPERIENCE

**Bioinformatics Analyst Intern – R&D** | *Elucidata Corporation, USA* July 2025 – October 2025

- Optimized AI-assisted NGS workflows by ~30% via prompt refinement and evaluation, improving biological reasoning for DEG and pathway analysis.
- Evaluated genomic workflow frameworks across 100+ curated research scenarios, ensuring biologically consistent and reproducible outputs before release.
- Performed LLM evaluation and prompt engineering, strengthening biological reasoning for KG, bulk/scRNA-seq, Synthea, and DEG tasks by 25–40%.
- Verified Synthea (synthetic health data) pipelines on cloud compute by confirming disease-specific cohort accuracy across 50+ repeatable test runs.
- Strengthened end-user research usability by stress-testing AI/ML agents across 100+ biological queries, improving genomics output quality & KG-driven insights.

**Research Assistant – Global Health Data Analytics** | *IU BioHealth Informatics Research Center, USA* September 2023 – May 2025

- Integrated team efforts to optimize RNA-seq preprocessing on HPC SLURM, cutting runtime by 40% using a Snakemake/Nextflow ready framework.
- Engineered reproducible RNA-seq pipelines using STAR, SAMtools, and BEDTools, generating high-quality, fully traceable gene-level feature matrices.
- Led QC & preprocessing automation using Bash scripting and SLURM batch jobs, improving pipeline stability by 25% & reducing manual team effort.
- Created pathway and network visualizations in Cytoscape, translating enrichment results into interpretable, biologically meaningful findings.

**QC Analyst** | *Biocon Biologics, India* July 2021 – August 2023

- Supported scRNA-seq clustering, annotation & biomarker analysis, improving insights by 25% and contributing to assay and preclinical decisions.
- Executed SOP approved workflows under GxP/GLP/ALCOA+, ensuring 100% reproducibility & reducing gaps by 40%.
- Collaborated with R&D, engineering, and IT data teams to resolve pipeline issues, boosting workflow scalability by 20% and accelerating review cycles.
- Worked hands-on with wet-lab scientists on PCR-based RNA extraction, NGS library preparation, and assay interpretation, delivering analyses that align with experimental goals and enable timely decision-making.

## ACADEMIC PROJECTS

**Spatial Transcriptomics of Human Breast Cancer** | Scanpy/Squidpy, 10x Genomics, Visium

- Engineered end-to-end ST pipeline on 4,869 spots × 21,349 genes; Leiden clustering resolved 11 TME domains with 99.4% spot retention.
- Detected Tertiary Lymphoid Structure (TLS) via Moran's I (2,486 SVGs, FDR<0.05), 24-signature deconvolution, and 39 LR pair mapping.
- Mapped dual PD-L1/TGF-β immunosuppressive barrier across 11 TME pathways; validated by 500-permutation testing revealing immune exclusion.

**Machine Learning Pipeline for Breast Cancer Biomarker Discovery** | Python (Pandas, NumPy, scikit-learn, Matplotlib)

- Harmonized 7 GEO cohorts (814 samples) via GPL-based probe-to-gene mapping to address cross-platform transcriptomic variability.
- Benchmarked five ML models with independent external validation; ANN achieved best performance (AUROC 0.911, specificity 0.988).
- Optimized a top-10 biomarker panel, improving external tumor detection by 17.9 percentage points (48.6% to 66.5%) while maintaining 99.4% specificity.

**WGCNA on COVID-19 and RSV Transcriptomes** | FastQC, HISAT2, SAMtools, R (Bioconductor, Tidyverse)

- Implemented an end-to-end RNA-seq pipeline (FastQC, HISAT2, featureCounts) on HPC, achieving >89% alignment across 68 samples.
- Applied WGCNA to identify key gene modules, showing strong correlation between MEyellow and COVID-19 severity (r = 0.89).
- Identified three shared antiviral hub genes (OASL, TXN, RBCK1) across COVID-19 and RSV datasets for downstream functional validation.

## EDUCATION

**Master of Science in Bioinformatics** August 2023 - May 2025

Luddy School of Informatics, Indiana University Indianapolis Indiana, USA

**Bachelor of Technology in Biotechnology** August 2016 - August 2020

Sapthagiri College of Engineering (Affiliated to VTU) Bangalore, India

## PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

**Biosciences Program Certification — Biocon Academy × KGI (2021)**

Industry training in biologics R&D, manufacturing, QC/QA, RA with hands-on lab, fermentation, mammalian cell culture, and drug-development projects.